



# RESPONSIBLE AI DASHBOARD FOR MODEL RISK ASSESSMENT

AI, Power & Responsibility – Governing Intelligent Systems for Trustworthy AI



Transparency



Fairness



Accountability



Governance



## 01 MOTIVATION

AI models are increasingly used in critical decision-making across finance, healthcare, insurance, recruitment and other sectors. However, these models can introduce risks related to bias, lack of transparency, performance degradation, drift, and regulatory non-compliance.



### Goal

Develop a Responsible AI Dashboard that continuously monitors model risk, fairness, explainability, drift and governance indicators to ensure trustworthy and accountable AI.



## DATASETS



### Cybersecurity Intrusion Dataset

- Cybersecurity Intrusion Detection Dataset
- Network traffic features
- Attack detection labels



### Sustainability / Environmental Dataset

- Sustainability & Environmental Indicators Dataset
- CO<sub>2</sub> Emissions
- Renewable Energy & Economic Indicators

All datasets are publicly available for research purposes.



## METHODOLOGY WORKFLOW

1

### Data Collection

Collect data from public sources and repositories.

2

### Data Cleaning & Preprocessing

Handle missing values, outliers and encode categorical variables.

3

### Feature Engineering

Create meaningful features to improve model performance.

4

### Model Training

Train ML models (Logistic Regression, XGBoost, Random Forest).

5

### Explainability Analysis

Use SHAP values to interpret model predictions.

6

### Fairness Assessment

Evaluate bias across demographic groups using fairness metrics.

7

### Drift Detection

Monitor data & concept drift over time to detect model degradation.

8

### Risk Scoring Engine

Aggregate metrics to compute overall risk score.

9

### Responsible AI Dashboard

Visualize metrics and alerts in real-time.

10

### Governance Recommendations

Provide actionable insights for governance and compliance.

## ? RESEARCH QUESTIONS

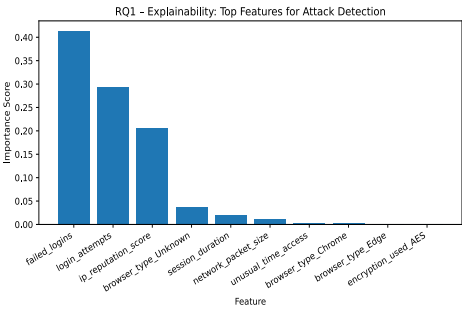
- RQ1** How explainable are AI model predictions?
- RQ2** Fairness assessment indicated low bias across network activity groups?
- RQ3** How stable is model performance over time?
- RQ4** Can model drift be detected automatically?
- RQ5** How can a dashboard improve AI governance and accountability?



## KEY RESULTS (HIGHLIGHTS)

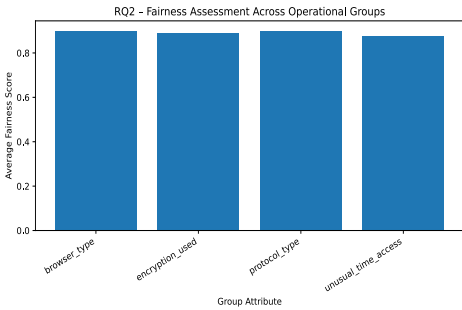
- SHAP analysis identified the most influential features driving intrusion detection predictions.
- Fairness assessment indicated low bias across network activity groups.
- The model achieved **88.47% accuracy** while maintaining stable prediction performance.
- Drift detection reported a minimal drift score of **0.0043**, indicating stable operational behavior.
- The Responsible AI Dashboard generated an **Overall Risk Score of 86/100**, supporting governance and compliance decisions.
- Continuous monitoring improved transparency, accountability, and trust in AI-driven decision making.

### RQ1 – Feature Importance (SHAP)



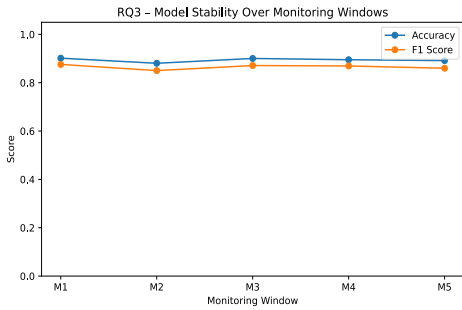
SHAP values show the impact of each feature on model predictions.

### RQ2 – Fairness Assessment



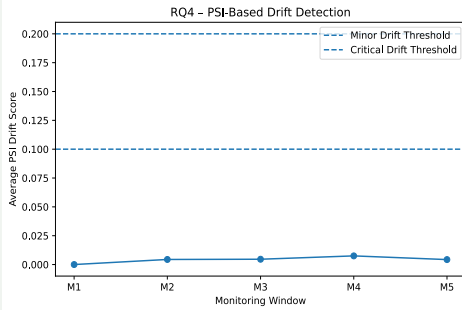
Fairness scores closer to 1 indicate lower bias across demographic groups.

### RQ3 – Model Performance Over Time



Slight performance degradation observed over time.

### RQ4 – Drift Detection



Drift score increases over time indicating distribution changes in the data.

### RQ5 – Risk Assessment Summary

86

Overall Risk Score  
(out of 100)



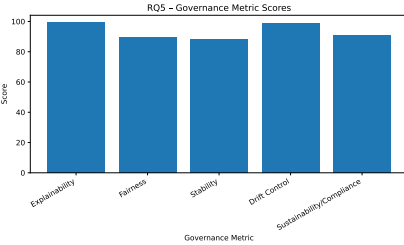
0.00 – 0.05: No Drift



0.05 – 0.15: Minor Drift

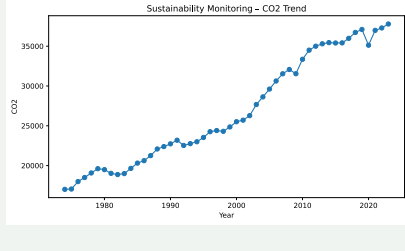


>0.15: Critical Drift



Aggregated risk score to support governance decisions.

| Metric               | Score  |
|----------------------|--------|
| Explainability Score | 99/100 |
| Fairness Score       | 89/100 |
| Stability Score      | 88/100 |
| Compliance Score     | 90/100 |
| Overall Risk Score   | 86/100 |



## DASHBOARD SNAPSHOT (EXAMPLE)

### Model Accuracy

88.47%

↑ 2% vs last month

### Fairness Score

89.42%

↑ 3% vs last month

### Drift Score

0.0043

Stable

### Explainability Score

99.14%

↑ 5% vs last month

### Compliance Status

PASS

All checks passed

### Overall Risk Score

86/100

Low Risk



### Alerts & Insights

• Drift score remains within acceptable threshold.

• Explainability analysis identified high-impact prediction features.

• Fairness metrics remain within governance limits.

• Continue periodic monitoring and model validation.

## CONTRIBUTIONS



Comprehensive Responsible AI Dashboard for model monitoring.



Integration of Explainability (SHAP) for transparency.



Bias and Fairness assessment across operational network groups.



Automated Drift Detection for continuous monitoring.



Risk Scoring Engine for actionable insights.



Supports Governance, Compliance and Accountability.



## 08 CONCLUSION

- The Responsible AI Dashboard provides a comprehensive framework for monitoring, evaluating, and governing AI systems throughout their lifecycle. By integrating explainability, fairness assessment, performance tracking, drift detection, and compliance monitoring, the dashboard enables organizations to identify risks before they impact decision-making processes.
- The results demonstrate that continuous monitoring significantly improves transparency, accountability, and trustworthiness in AI deployments. The unified risk scoring mechanism simplifies complex model evaluation metrics into actionable insights for stakeholders, regulators, and decision-makers.
- Overall, this work highlights the importance of Responsible AI practices and demonstrates how governance-driven dashboards can support ethical, fair, and reliable AI systems in real-world applications.
- The proposed framework achieved an **Explainability Score of 99.14%**, **Fairness Score of 89.42%**, and **Overall Risk Score of 86/100** while maintaining a minimal **drift score of 0.0043**.
- These results demonstrate the effectiveness of continuous Responsible AI monitoring and governance.



Trustworthy AI



Fair & Unbiased



Transparent



Accountable

